

NITIN SINGH

(+91) 7041808600 | Surat, India

ni3.singh.r@gmail.com | www.linkedin.com/in/nitinsinghr/ | www.github.com/NI3singh | [Portfolio](#)

Professional Summary

AI/ML Engineer with 2 years of experience in developing LLM-powered applications, RAG pipelines, and intelligent workflows using Python, FastAPI, LangChain. Specialized in Agentic AI workflows. Skilled in fine-tuning models (LoRA), **optimizing inference pipelines for cost saving**, building multi-agent systems for automated decision-making, and deploying scalable solutions on AWS.

Education

Uka Tarsadia University | Surat, Gujarat

B.Tech Artificial Intelligence & Data Science

Achieved a perfect 10.0 SGPA in the final semester, demonstrating mastery of advanced coursework.

CGPA: 8.68

Sep 2021 - May 2025

Publications

Is Architectural Complexity Always the Answer? A Case Study on SwinIR vs. an Efficient CNN | Chandresh Sutariya, Nitin Singh

- Abstract:** It has been demonstrated that an optimized CNN achieves near state-of-the-art restoration quality, comparable to SwinIR, while significantly reducing computational overhead.
- [arXiv Pre-print, 2025](#)

Skills

Deep Learning & CV : PyTorch, TensorFlow, LoRA Fine-Tuning, Stable Diffusion, Computer Vision, Inference Optimization, Supervised Fine-Tuning (SFT)

Generative AI & LLMs : OpenAI API, Google Gemini, LangChain, RAG Pipelines, MCP, Agentic AI, ADK, SDK, Multi-Agent Systems, LLM Evaluation, Prompt Engineering

Languages & Databases : Python (Advanced), React.js, PostgreSQL, Vector Databases (ChromaDB/Pinecone/FAISS), MongoDB

Cloud & Tools : Amazon Web Services (AWS), Docker, Git, Streamlit, Flask, FastAPI, Kohya_ss, ComfyUI

Soft Skills : Communication, Discipline, Problem Solving, Leadership, Consistency, Proactiveness, Reliability, AI Enthusiast

Projects

AI Resume Builder (ResumeTeX) | FastAPI, Next.js 14, PostgreSQL, Python, pytest

[Github Link](#) | [Live Link](#)

- Built a deterministic anti-fabrication guard in Python for an LLM resume-tailoring pipeline, validated on 5 real resumes using MiniMaxAI/MiniM2.5, where the production prompt produced ~4 fabricated claims per transform (~20 under a naive baseline) and the guard eliminated 100% in both conditions, reverting up to 34 immutable-field edits.
- Designed a 27-case adversarial benchmark confirming 100% fabrication recall, reduced the guard's false-positive rate from 56% to 33% by implementing a spelled-out-cardinal normalizer, and authored a pytest suite of 46 unit tests + 8 xfail blind-spot characterizations achieving 100% line coverage on the guard module.
- Benchmarked the 3-stage import pipeline on 9 real resumes, extracting 70 hyperlinks (77% invisible to plain-text extraction) by parsing PDF link annotations and DOCX relationships, and measured URL-to-field routing at 85% accuracy with 100% accuracy on contact, profile, credential, and website links.
- Shipped a full-stack AI resume builder using Next.js 14, FastAPI, and PostgreSQL JSONB with live LaTeX generation, one-click PDF compilation, JWT HttpOnly auth with server-side session revocation, and unlimited resume branching.

StockStalker-AI (Multi-agent Financial Intelligence) | Gen AI, RAG, FastMCP, PostgreSQL

[Github Link](#) | [Live Link](#)

- Built a financial news engine that aggregates and deduplicates articles from three sources (TradingView, Finviz, Polygon, SEC EDGAR), using Gemini Pro to select the 5–7 most relevant and produce concise (500-word) summaries grounded in the source text.
- Engineered a provider-agnostic LLM factory using Pydantic v2 for end-to-end typed pipeline validation, with generate structured() to eliminate fragile JSON parsing and markdown-stripping hacks
- Developed quantitative analysis agent computing technical signals (RSI, MACD, Bollinger Bands, volume ratio) from 60-day yfinance data using pandas ta, with isolated indicator calculations so single failures degrade gracefully

Generative AI Image Pipeline & LoRA Fine-Tuning | Deep Learning & Computer Vision

[Github Link](#)

- Built AI-Avtaar, a Streamlit app orchestrating a full character-creation pipeline across three SDXL engines: LoRA training (Kohya_ss), image generation (Automatic1111), and virtual try-on (CatVTON).
- Automated the training flow end-to-end auto-captioning, dataset preparation, and a zero-config launch that trains a custom character LoRA in 30–60 minutes with live log streaming.
- Integrated CatVTON so users overlay arbitrary clothing onto generated avatars, running each engine in an isolated virtual environment to prevent dependency conflicts.

BTC-Forecaster (Multi-Agent Intraday Forecasting System) | Agentic AI & LLM Orchestration

[Github Link](#)

- Forked and re-engineered TradingAgents, an open-source multi-agent LangGraph framework built for multi-week stock investment decisions, into an intraday BTC-USD price forecaster predicting direction, price, and range over 1-hour and 4-hour horizons.
- Redesigned the output schema, prompts, and data pipeline to run on hourly OHLCV bars instead of daily bars, while preserving the existing 12-role agent pipeline, including technical, sentiment, and news analysts, a bull/bear research debate, and a 3-way risk team.
- Built a new forecast evaluation system that logs every prediction and automatically grades it against the realized price once its time horizon elapses, tracking directional accuracy, range coverage, and price error over time.

Experience

AI/ML Engineer ELaunch Solution Pvt. Ltd., Surat, Gujarat	Dec 2024 - Jun 2026
- Worked on an AI-driven game-personalization-engine utilizing Behavioral Session Telemetry and reinforcement learning (RTP, game volatility) to maximize player engagement in real-time. - Engineered an end-to-end data pipeline for Solana price analysis, ingesting OHLCV data from Binance API spanning 01-01-2021 to 2909-2024 (1300+ days) and performing data cleaning, merging, transformation, feature engineering to generate a 44-column feature-rich dataset by adding technical indicators (RSI, MACD, Bollinger Bands, moving averages). - Trained a Temporal Fusion Transformer (TFT) model using PyTorch with GPU acceleration for Solana price prediction, achieving a validation loss of 0.7376 (epoch 34). Evaluated model performance on test data using MAE, RMSE, and MAPE metrics. - Built and deployed 3 production-ready Naive RAG and Agentic RAG pipelines, integrating the ElevenLabs Voice API for real-time voice interactions, including AI chatbots capable of querying 10,000+ documents and enterprise MongoDB databases using semantic search and LLM-powered retrieval.	
Data Analyst Intern JBcodeapp, Surat, Gujarat	May 2024 - Jun 2024
<i>Project: Student Performance Analysis — Python, NumPy</i> - Engineered an automated data pipeline using Python to perform ETL (Extract, Transform, Load) on 5,000+ student records. - Built a scikit-learn clustering pipeline using Agglomerative Clustering (n=3 clusters) and Linear Regression for trend analysis to categorize students into performance groups (Strong, Average, Weak) based on longitudinal exam scores. - Synthesized findings into a comprehensive report that isolated key factors influencing academic scores, providing a foundation for data-driven teaching methods and resource allocation strategies.	
AI/ML Trainee AlgoBrain AI, Surat, Gujarat	Sep 2023 - Jan 2024
<i>Project: Movie-Ticket Booking Chatbot with DialogFlow CX</i> Analyzed potential user conversation paths to design and build a movie-ticket booking chatbot in Dialogflow CX, enhancing the overall user experience.	

Extracurricular Activities

Research Community Member Swaayan Drone Club, SVNIT, Surat, Gujarat	May 2024 - Jun 2024
Built and Assisted with Drone Bootcamps by building and calibrating drones using Betaflight and Mission Planner.	
User Software Tester (Part-Time) Nothing India, Remote	Feb 2023 - Present
Conducted beta testing for Nothing Phone 1, Nothing Phone 2, Nothing 2a, Android 14, CMF Watch, and more.	
Aspire Leaders Program Aspire Institute, Harvard Business School, Online	Aug 2024 - Nov 2024
Completed all modules in Leadership, Communication, and Problem-Solving.	
Google Student Ambassador Program Google, Ping Network MCN, Online	Sep 2025 - Dec 2025
Selected as the Google Student Ambassador from my college, serving as the campus representative for Google.	
Minimax Ambassador Minimax.io, Remote	Jan 2026 - Present
Selected as Minimax Ambassador to test the model and give feedback.	
Gujarat State Lead Open Source Connect, Remote	Jul 2025 - Sep 2025
Led Open Source Connect initiatives in Gujarat, driving AI, ML, and marketing outreach efforts.	

Achievements

- Awarded a Certificate of Excellence in Summer Analytics (IIT Guwahati) for finishing in the top 10 of all participants. - Certified in Time Series Analysis by IIT Guwahati, with demonstrated expertise in statistical forecasting models. - Achieved Runner-Up position in Tattva Hackathon 2025 (CGPIT), demonstrating problem-solving, teamwork, and innovation. - Achieved Winner position in Quiz (Group) Competition at College Fest 2023, showcasing knowledge, teamwork, and quick thinking. - Secured 2nd Runner-Up position in Script Writing (Solo) at College Fest 2023, demonstrating creativity and communication skills. - Secured Top 5 finishes at IITGN (Finnovate/Amalthea) and IIT Bombay (AMD) Hackathons, recognized for technical excellence and innovative problem-solving.	
--	--

Certifications

- Oracle AI Vector Search Certified Professional [View Credential] - Natural Language Processing in TensorFlow [View Credential] - Retrieval Augmented Generation (RAG) [View Credential] - Fine Tune BERT for Text Classification with TensorFlow [View Credential] - Machine Learning Specialization [View Credential] - Completed ML: An Overview from Coursera [View Credential]	
--	--